```
ArmourStone Assessment Report
===========================
Generated: 2026-05-20 02:59:29

Ship
----
Propeller diameter (D_p) [m]: 1.0000
Propeller vertical position (Z_p) [m]: 5.0000
Hull draught [m]: 7
Jet velocity [m/s]: 3.0000

Waterway
--------
Water depth h [m]: 10.0000
Propeller-to-slope distance [m]: 5.0000
Slope angle beta [deg]: 30.0000
Armourstone repose angle phi_as [deg]: 45.0000
Flow direction psi_flow [deg]: 90

ArmourStone
-----------
Rock density [kg/m^3]: 2650
Water density [kg/m^3]: 1025
Critical mobility psi_cr [-]: 0.0350
Stability correction phi_sc [-]: 0.7500
Turbulence factor k_t2 [-]: 4
Roughness height k_s [m]: 0.3000
Manual hydraulic velocity [m/s]: 2.0000

CFD
---
Use CFD: True
Simulation type: 2D
Left boundary to propeller 2D [m]: 20.0000
Left boundary to propeller 3D [m]: 1.0000
Domain width 3D [m]: 12.0000
Cells flat x 2D: 240
Cells slope x 2D: 80
Cells z 2D: 80
Cells flat x 3D: 180
Cells slope x 3D: 60
Cells y 3D: 36
Cells z 3D: 50

Derived parameters
------------------
Relative buoyant density delta [-]: 1.5854
Velocity profile factor k_h [-]: 0.0557
Side slope factor k_sl [-]: 0.7071
Hydraulic velocity [m/s]: 1.9785
Required D_n50 [m]: 0.4656
Required D_n50 [cm]: 46.5566

CFD results
-----------
Latest time [s]: 3470
Max flat bed velocity [m/s]: 1.9785
Max slope velocity [m/s]: 0.3555
Governing velocity [m/s]: 1.9785

Summary
-------
Required D_n50 [m]: 0.4656
Required D_n50 [cm]: 46.5566

For better visualization
------------------------
The .foam file generated by the CFD workflow can be found in the following directory:: cases/case_Test_1_2D/case_Test_1_2D.foam
This file can be opened in ParaView or other applications that support OpenFOAM files to visualize the CFD results in more detail.:
```